

Frequency-specific Sensory Stimulation Reduces Stress Potentiated Deficits In 5xfad Mice

Tina Carla Franklin¹, Hayden Moore¹, Krithi Iyer¹, Abigail Copeland¹, Matthew Goodson¹, Jais Matta¹, Ashley Prichard, PhD², Annabelle Catherine Singer¹.

¹Biomedical Engineering, Georgia Institute of Technology, Atlanta, GA, USA, ²Veterans Affairs Medical Center, Durham, NC, USA.

Background: Severe or chronic stress drastically increases Alzheimer's disease risk, likely through stress-induced disruption of neuroimmune function and the gut microbiome. We and others previously demonstrated that noninvasive, periodic audiovisual stimulation at specific frequencies, called "flicker," modulates neuroimmune signaling in brain regions implicated in stress-related and neurodegenerative disorders. In unstressed mouse models of amyloidosis, 40 Hz flicker improves microglial and neuronal function. Separately, we have shown that frequency-specific flicker attenuates stress-induced behavioral, synaptic, and inflammatory phenotypes in a sex-dependent manner. However, the combined impact of chronic stress and amyloid pathology, a clinically relevant multi-hit condition in which multiple risk factors converge to accelerate disease progression, remains poorly understood, and whether flicker can mitigate stress-potentiated deficits in this context is unknown.

Methods: To address this gap, we characterized behavioral, cellular, and gut microbiome effects of chronic stress in the 5XFAD amyloidosis mouse model and determined whether a noninvasive intervention enhances stress resilience at frequencies previously shown to support brain health. Male and female 5xFAD mice were exposed to 4 weeks of chronic unpredictable stress (CUS) and concomitantly treated with audiovisual flicker at 10 Hz or 40 Hz, frequencies previously shown to ameliorate stress- or amyloid-associated pathology in isolation. We assessed stress and flicker effects on physiology, behavior, amyloid pathology, microglia, and the gut microbiome. We quantified amyloid deposition and microglia morphology in the hippocampus following 3D rendering.

Results: CUS significantly exacerbated physiological stress responses in 5XFAD mice, indicated by locomotor and anxiety-like behavioral impairments. Flicker mitigated these deficits in a frequency- and sex-dependent manner, with the strongest protection in males. We found that stress exacerbated amyloid deposition and promoted accumulation of large plaques with reduced microglia contact. Frequency-specific flicker prevented stress-potentiated plaque accumulation at frequencies previously shown to promote resilience. Flicker that prevented plaque accumulation improved microglial colocalization and engulfment of amyloid. Lastly, 16S rRNA sequencing revealed sex-specific, stress-associated fecal microbiome signatures in 5XFAD mice that were mitigated by chronic flicker exposure.

Conclusion: These findings identify chronic stress as a potent amplifier of neurodegenerative deficits and show that frequency-specific flicker counteracts the combined harm of stress and amyloid.